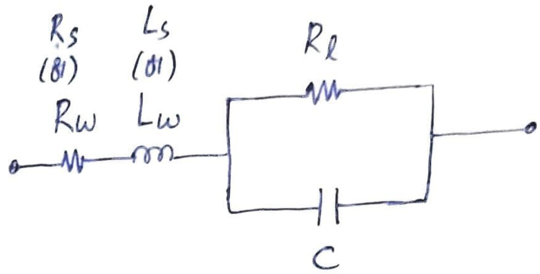
EE6203 : ASSIGNMENT - 4

(Q1)

(1)



R_s / R_w = resistance due to leads / wires.

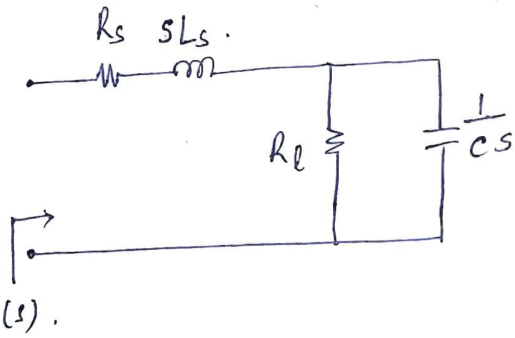
L_s / L_w = inductance due to leads / wires.

R_l = insulation leakage resistance.

C = capacitance.

(2)

$$Z(s) = R_s + sL_s + \frac{R_l \cdot \frac{1}{sC}}{R_l + \frac{1}{sC}}$$



Simplifying it results as,

$$Z(s) = \frac{(R_s + R_l)\left[s^2 \left(\frac{L_s R_l C}{R_s + R_l} \right) + s \left(\frac{L_s + R_s R_l C}{R_s + R_l} \right) + 1 \right]}{1 + \frac{s}{R_l C}}$$

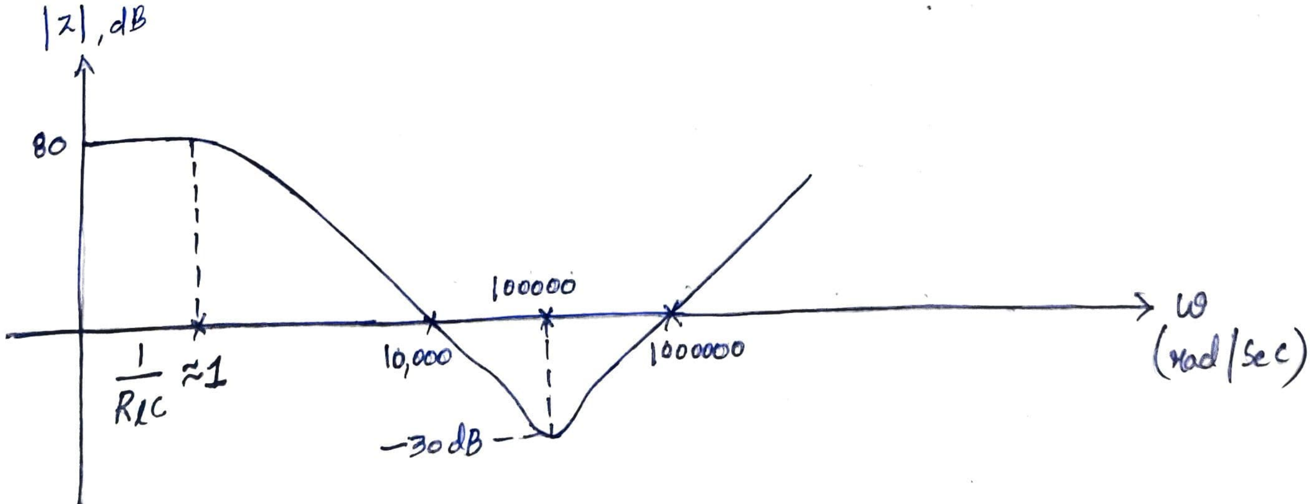


Fig (1)

Magnitude plot of $Z(s)$ is shown in Fig (1).

$$\omega < \frac{1}{R_\ell C} \quad \Rightarrow \quad |Z| = R_s + R_\ell$$

$$|Z| \simeq R_\ell \quad (\because R_\ell >> R_s).$$

$$\therefore \quad 20 \log |Z| = 80$$

$$|Z| = 10^4$$

$$\boxed{\therefore \; R_\ell = 10 \text{ k}\Omega}$$

From the plot $\frac{1}{R_\ell C} \approx 1$

$$\therefore \quad C = \frac{1}{R_\ell} = \frac{1}{10{,}000}$$

$$\boxed{\therefore \; C = 100 \, \mu F}$$

Fig (2) shows the standard model of the capacitor, where

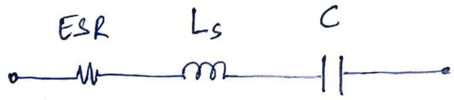


Fig (2)

$$ESR = R_s + \frac{1}{\omega^2 R_\ell C^2} \longrightarrow (1)$$

From Fig (1) at $\omega = 10^5$ rad/sec $|Z| = ESR$.

$$\therefore \quad 20 \log |Z| = -30$$

$$|Z| = 0.0316$$

$$\therefore \quad \boxed{ESR = 31.6\ m\Omega}$$

From ① $\quad R_s + \dfrac{1}{\omega^2 R_\ell C^2} = 31.6 \times 10^{-3}$

$$R_s + \frac{1}{(10^5)^2 \times 10 \times 10^3 \times (100 \times 10^{-6})^2} = 31.6 \times 10^{-3}$$

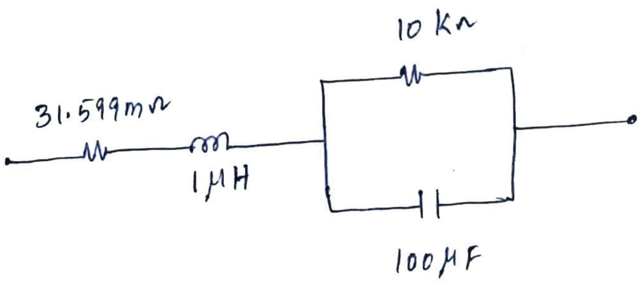
$$\boxed{R_s = 31.599\ m\Omega}$$

From Fig (1) at $\omega = 10^5$ rad/sec, network behaves as pure resistive.

$$\therefore \quad \omega = \frac{1}{\sqrt{L_s C}} \Rightarrow 10^5 = \frac{1}{\sqrt{L_s \cdot 100 \times 10^{-6}}}$$

$$\boxed{\therefore \quad L_s = 1\ \mu H}$$

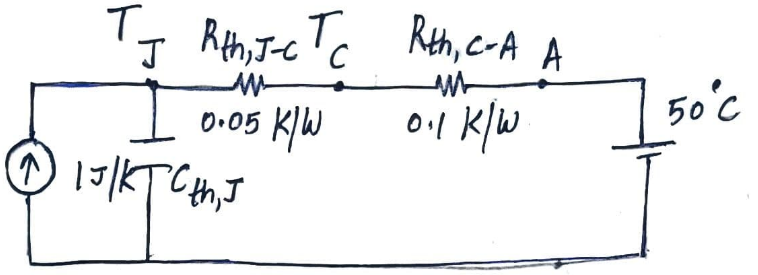
$\therefore$



(Q 2)

Given

$P_D = 500W$



(A) To find steady state temperatures of junction and case.

Case temperature T_C:

$$P_D = \frac{T_C - T_A}{R_{th,C-A}}$$

$$500 = \frac{T_C - 50}{0.1}$$

$$\boxed{\therefore T_C : 50°C}$$

Junction temperature T_J :-

$$P_D = \frac{T_J - T_C}{R_{th,J-C}} \Rightarrow 500 = \frac{T_J - 50}{0.05}$$

$$\boxed{\therefore T_J = 125°C}$$

(B) Transient thermal impedance, $Z_{th} = R_{th}\left(1 - e^{-t/\tau}\right) = 0.15\left(1 - e^{-t/0.15}\right)$.

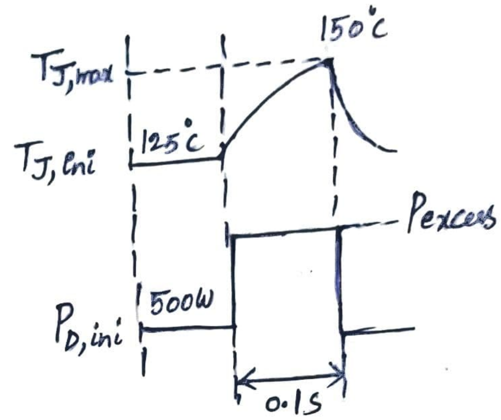
$$\tau = \left(R_{th,J-C} + R_{th,C-A}\right) C = (0.05 + 0.1)(1) = 0.15$$

$$\therefore Z_{th}\Big|_{t=0.1s} = (0.15)\left[1 - e^{-0.1/0.15}\right]$$

$$\boxed{Z_{th}\Big|_{t=0.1s} = 0.073 \text{ K/W}}$$

(c) $\Delta T_J = P_{excess} Z_{th}$

$$P_{excess} = \frac{T_{J,max} - T_{J,ini}}{Z_{th}}$$



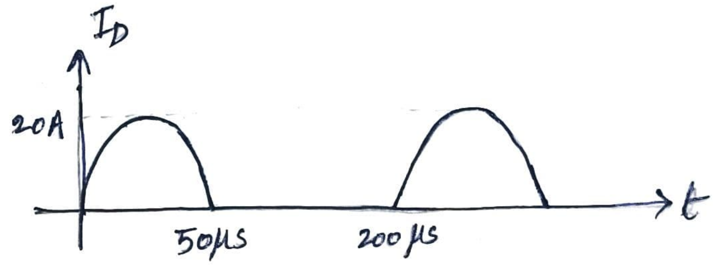
$$= \frac{150^\circ - 125^\circ}{0.073}$$

$$\boxed{P_{excess} = 342.465 \text{ W}}$$

(A)

(Q3) Device - D :-

Average calculation :- $I_{D,avg}$.



$$I_{D,avg} = \frac{1}{T}\int_0^T I_m \sin\omega t \, dt$$

$$= \frac{1}{T}\int_0^t I_m \sin\omega t \, dt = \frac{1}{200\times10^{-6}}\int_0^{50\times10^{-6}} 20 \sin\omega t \, dt$$

$$= \frac{20}{200\times10^{-6}} \times \frac{1}{\omega}\left[-\cos\omega t\right]_0^t$$

$$= \frac{20}{200\times10^{-6}\times 2\times\pi\times5000}\left[-\cos\left(2\times\pi\times5000\times50\times10^{-6}\right) + 1\right]$$

$$\boxed{\therefore\ I_{D,avg} = 3.183 \text{ A}}$$

RMS Calculation : $I_{D,rms}$

$$I_{D,rms}^2 = \frac{1}{T}\int_0^t I_m^2 \sin^2 wt \, dt$$

$$= \frac{I_m^2}{T}\int_0^t \frac{1-\cos 2wt}{2} dt$$

$$= \frac{I_m^2}{2T}\left[\int_0^t dt - \int_0^t \cos 2wt \, dt\right]$$

$$= \frac{I_m^2}{2T}\left[t - \frac{1}{2w}(\sin 2wt)_0^t\right]$$

$$= \frac{I_m^2}{2T}\left[t - \frac{\sin 2wt}{2w}\right]$$

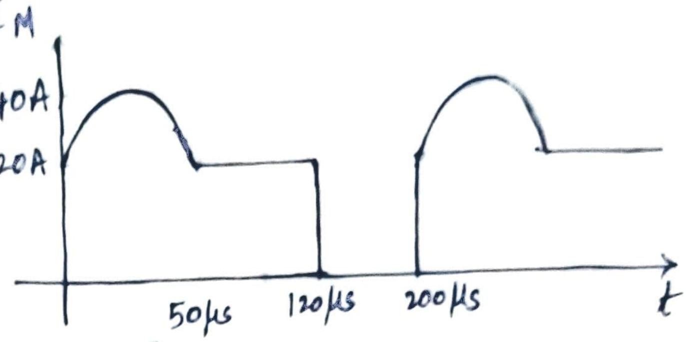
$$= \frac{(20)^2}{2\times 200\times 10^{-6}}\left[50\times 10^{-6} - \frac{\sin(2\times 2\times \pi\times 5000\times 50\times 10^{-6})}{2\times 2\times \pi\times 5000}\right]$$

$$I_{D,rms}^2 = 50$$

$$\boxed{\therefore \; I_{D,rms} = 7.071 \text{ A}}$$

Device M:

Average Calculation $I_{M,avg}$



$$I_{M,avg} = \frac{1}{T}\left[\int_0^{50\times10^{-6}} (20 + 20\sin\omega t)\,dt + \int_{50\times10^{-6}}^{120\times10^{-6}} 20\,dt\right]$$

$$= \frac{1}{T}\left[\int_0^{50\times10^{-6}} 20\,dt + \int_0^{50\times10^{-6}} 20\sin\omega t\,dt + \int_{50\times10^{-6}}^{120\times10^{-6}} 20\,dt\right] \qquad T = 200\times10^{-6}$$

Solving the above integration results in

$$\boxed{I_{M,avg} = 15.183\ A}$$

RMS Calculation : $I_{M,rms}$

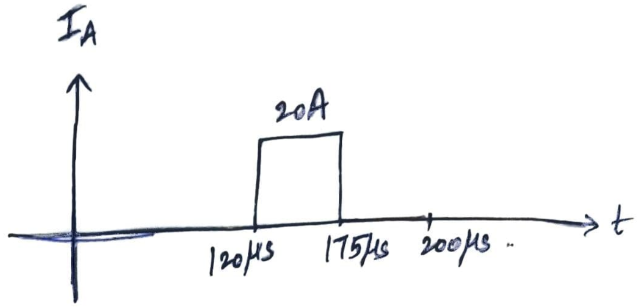
$$I^2_{M,rms} = \frac{1}{T}\left[\int_0^{50\times10^{-6}} (20 + 20\sin\omega t)^2\,dt + \int_{50\times10^{-6}}^{120\times10^{-6}} (20)^2\,dt\right]$$

$$= \frac{1}{200\times10^{-6}}\left[\int_0^{50\times10^{-6}} 400\,dt + \int_0^{50\times10^{-6}} 400\sin^2\omega t\,dt + \int_0^{50\times10^{-6}} 800\sin\omega t\,dt + \int_{50\times10^{-6}}^{120\times10^{-6}} 400\,dt\right]$$

Solving the above integration

$$\boxed{I_{M,rms} = 20.37\ A}$$

Device A :-



Average Calculation: $I_{A,avg}$

$$I_{A,avg} = \frac{1}{T}\int_0^t i_A(t)\,dt$$

$$= \frac{1}{200\times10^{-6}}\left[\int_{120\times10^{-6}}^{175\times10^{-6}} 20\,dt + \int_{175\times10^{-6}}^{200\times10^{-6}} 0\,dt\right]$$

$$\boxed{\therefore\ I_{A,avg} = 5.5\ A}$$

RMS Calculation: $I_{A,rms}$

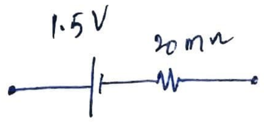
$$I_{A,rms}^2 = \frac{1}{T}\int_0^t i_A^2(t)\,dt$$

$$= \frac{1}{200\times10^{-6}}\int_{120\times10^{-6}}^{175\times10^{-6}} (20)^2\,dt$$

$$\boxed{I_{A,rms} = 10.488\ A}$$

(B) To calculate average power loss :-

Device M:-

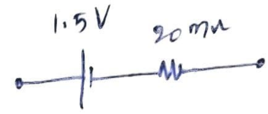


$$P_M = I_{M,avg}(1.5) + I_{M,rms}^2(20\times10^{-3})$$

$$= (15.183)(1.5) + (20.37)^2(20\times10^{-3})$$

$$\boxed{\therefore\ P_M = 31.072\ W}$$

Device A:-

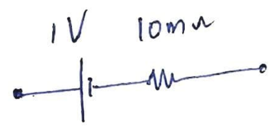


$$P_A = I_{A,avg}\,(1.5) + I_{A,rms}^{2}\,(20\times10^{-3})$$

$$= (5.5)(1.5) + (10.488)^2\,(20\times10^{-3})$$

$$\boxed{P_A = 10.449\ W}$$

Device D



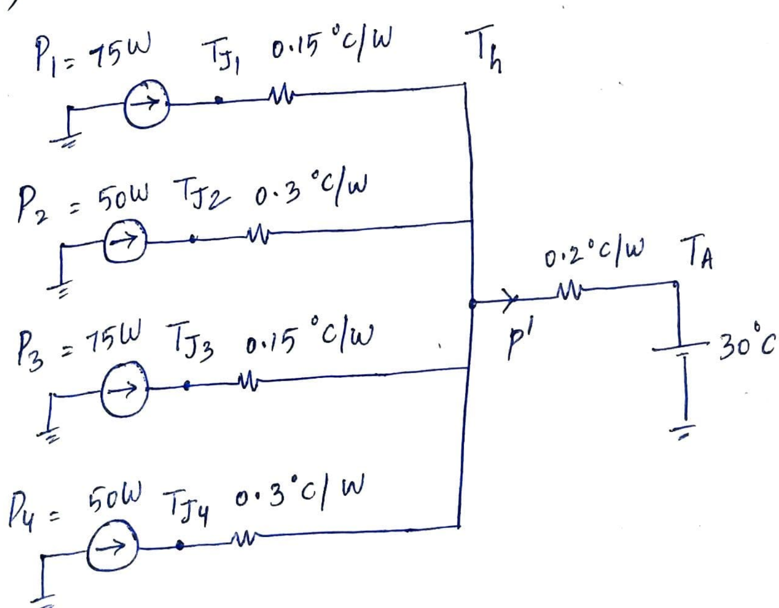
$$P_D = I_{D,avg}\,(1) + I_{D,rms}^{2}\,(10\times10^{-3})$$

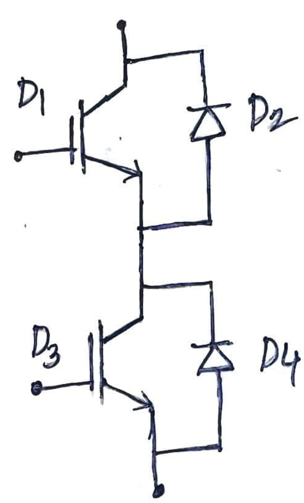
$$= (3.183)(1) + (7.071)^2\,(10\times10^{-3})$$

$$\boxed{P_D = 3.6829\ W}$$

(Q4)

(A)





(B) To find heat sink temperature :-

From above figure $P' = P_1 + P_2 + P_3 + P_4 = 250W$

$$P' = \frac{T_h - T_A}{R_{th,h-A}}$$

$$250 = \frac{T_h - 30}{0.2}$$

$$\boxed{T_h = 80^\circ C}$$

To find junction temperatures :-

Device - 1 :-

$$P_1 = \frac{T_{J_1} - T_h}{R_{th,J-h}} \Rightarrow 75 = \frac{T_{J_1} - 80}{0.15}$$

$$\boxed{\therefore T_{J_1} = 91.25^\circ C}$$

Device - 2

$$P_2 = \frac{T_{J_2} - T_h}{R_{th,J-h}} \Rightarrow \frac{T_{J_2} - 80}{0.3} = 50$$

$$\boxed{T_{J_2} = 95^\circ C}$$

Device - 3

$$P_3 = \frac{T_{J_3} - T_h}{R_{th,J-h}} \Rightarrow 75 = \frac{T_{J_3} - 80}{0.15}$$

$$\boxed{\therefore T_{J_3} = 91.25^\circ C}$$

Device - 4 :-

$$P_4 = \frac{T_{J_4} - T_h}{R_{th,J-h}} \Rightarrow 50 = \frac{T_{J_4} - 80}{0.3}$$

$$\boxed{T_{J_4} = 95^\circ C}$$